Visualization and Analysis Plan for DICE + Lenton SCM Integration

This document outlines data visualization and analysis strategies to present your integration of the DICE economic model with the Lenton (2000) Simple Climate Model (SCM), including permafrost feedbacks. The goal is to showcase the system-level implications of including climate–carbon feedbacks within DICE's optimal control framework — with the systems-thinking style favored by Brian Beckage.

1. Core Comparative Plots: Baseline vs. Coupled System

Goal: Show how adding the SCM and permafrost module changes DICE's optimal trajectories.

- Optimal mitigation rate (μ) over time — compare DICE default vs. DICE+SCM trajectories.
- Global temperature trajectories (TATM / Anomaly).
- Atmospheric carbon or radiative forcing.
- Social welfare, consumption, or damages.

Brian will appreciate clear annotation of divergence points and shaded regions highlighting feedback impacts.

2. Feedback Structure Visualization

Goal: Provide conceptual clarity through a causal loop diagram linking economy, climate, and carbon feedback subsystems.

Include: economic (DICE) subsystem, climate-carbon (SCM) subsystem, and cross-system feedback couplings like permafrost and vegetation carbon loops.

3. Sensitivity or Regime Analysis

Goal: Show robustness of optimal trajectories to feedback uncertainty.

- Parameter sweep on feedback strength or equilibrium climate sensitivity.
- Heatmaps or ribbons for $\mu(t)$ trajectories.
- Phase plots of feedback strength vs. long-term temperature or welfare.
- Identify thresholds where optimal control behavior shifts.

4. Feedback Decomposition

Goal: Quantify how much the added feedbacks shift system outputs.

Compute ΔT = T_DICE+SCM – T_DICE and attribute differences to permafrost carbon, ocean uptake, etc. Express extra feedback carbon as a percent of human emissions.

5. Emergent Properties and Trade-Offs

Goal: Explore system-level trade-offs that reveal emergent dynamics.

- Peak temperature vs. total mitigation cost.
- Net benefit vs. tipping feedback timing.
- Safe operating space diagrams (T vs. feedback strength).

6. Next Steps for Analysis

- Endogenize tipping awareness in the damage or mitigation functions.
- Test delayed policy responses (tipping ignorance vs. foresight).
- Assess robustness of optimal control under uncertainty.

7. Presentation Tips

- Use consistent time axes and color coding (e.g., blue = DICE, red = DICE+SCM).
- Annotate turning points and nonlinear feedback emergence.
- Open with a system map, then present quantitative comparisons.

These visualizations and analyses will demonstrate strong systems-thinking and analytical integration of climate-carbon feedbacks into economic decision dynamics.